



# Antidepressant-like effects of water extract of *Gastrodia elata* Blume in rats exposed to unpredictable chronic mild stress via modulation of monoamine regulatory pathways



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## ABSTRACT

**Ethnopharmacology relevance:** *Gastrodia elata* Blume (GE) is a traditional herbal medicine belonging to the *Orchidaceae* family, and has been used to manage neurological disorders for centuries. We have previously reported that its water extract (WGE) could improve the depressive-like behaviours in the forced swimming test (FST), an animal model of depression.

**Aim of the study:** To investigate the antidepressant-like effects of WGE in rats exposed to unpredictable chronic mild stress (UCMS) model, and to explore its possible molecular mechanisms.

**Materials and methods:** UCMS rats were orally administered with WGE (0.5 g/kg body weight) daily within the 4 weeks UCMS procedure. The sucrose preference test and the open field test were conducted to assess anhedonia and spontaneous behaviours, respectively. The cerebral turnover rates of monoamine neurotransmitters and the serum corticosterone levels were measured. *In vitro* direct and indirect monoamine oxidase A (MAO-A) inhibitory assays were employed to assess the possible antidepressant-like mechanisms of WGE (0.5 mg/mL) and its major component, gastrodin (GAS, 15, 30 and 60 µg/mL). Western blot was used to examine the expression of protein related to monoamine regulation, such as MAO-A and tyrosine hydroxylase (TH).

**Results:** WGE significantly reversed the sucrose preference and other abnormal behaviours induced by 4 weeks of UCMS. WGE significantly restored the cerebral turnover rates of serotonin and dopamine and decreased serum corticosterone levels. WGE and gastrodin inhibited the activity and protein expression of MAO-A, and increased TH levels in PC12 cells.

**Conclusion:** The antidepressant-like effects of WGE and gastrodin might be mediated by the regulation of monoamine neurotransmitters, and therefore were beneficial in depression treatment as a complementary approach.

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## 1. Introduction

The World Health Organization (WHO) predicted that depression will be the most important contributor to the global burden of disease and disability by 2030 (Mathers et al., 2008). Since the

treatments for depression are time-consuming (Nestler, 1998) and have undesirable side-effects, adherence to conventional medication against depression is poor (Ueda et al., 2003). Therefore, alternative approaches or complementary therapies, such as herbal remedies, should be investigated to prevent depression and to assist the currently used medications.

In recent decades, several studies supported that the monoamine neurotransmitter deficiency in neurons might be one of the major reasons for depression (Belmaker and Agam, 2008; Castren, 2005). The three key monoamine neurotransmitters are serotonin (5-hydroxytryptophan, 5-HT), norepinephrine (NE) and dopamine (DA), which are involved the regulation of mental status (Lanni et al., 2009). Low concentrations and high turnover rates of these neurotransmitters were found in the cerebrum of subjects with depression. Monoamine oxidases (MAO) enzymes are responsible

**Abbreviations:** WGE, water extract of *Gastrodia elata* Blume; GAS, gastrodin; UCMS, unpredictable chronic mild stress; MAO, monoamine oxidase; 5-HT, 5-hydroxytryptophan (serotonin); 5-HIAA, 5-hydroxyindoleacetic acid; DA, dopamine; DOPAC, 3,4-dihydroxyphenylacetic acid; HVA, homovanillic acid; PC12, pheochromocytoma cells; TH, tyrosine hydroxylase; CTL, control; CW, control plus WGE treatment; UW, UCMS plus WGE treatment

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for degrading monoamine neurotransmitters (Youdim et al., 2006), and there are two isoforms of MAO termed as MAO-A and MAO-B. MAO-A primary deaminates 5-HT, NE and DA. Elevation of MAO-A's activity is strongly associated with depression (Meyer et al., 2009; Shih et al., 1999). Antidepressants including tricyclic agents (TCAs), selective serotonin reuptake inhibitors (SSRIs) and MAO inhibitors (MAOIs) have therapeutic effects on depression.

Unpredictable chronic mild stress (UCMS) having a good ecological validity in mimicking the pathological progression of depression in human has been used to develop an animal model to study depression (Hill et al., 2012; Willner, 1997). Anhedonia is one of the core symptoms of depression in humans, and reduction in sucrose preference can be observed in UCMS model (Willner, 1997). Several abnormal behavioural phenotypes such as reduction of daily intakes, growth retardation and locomotor deficits are found in UCMS model. UCMS interferes with the normal circuit of reward system (Willner, 1997) including the prefrontal cortex, hippocampus, striatum and amygdala. In addition, chronic stress increases the activity of MAO-A (Lin et al., 2005), which may affect the utilization efficiency of monoamine neurotransmitters.

The PC12 rat pheochromocytoma cell line is derived from the adrenal medulla. Undifferentiated PC12 cells not only synthesize and store large amounts of DA, but also express enzymes such as tyrosine hydroxylase (TH) and MAO-A/B necessary for complete DA metabolism (Greene and Rein, 1977; Kishimoto et al., 2005). Therefore, undifferentiated PC12 cells are extensively used as an *in vitro* model for DA metabolism, investigations of neuroprotective effect and antidepressant-like activity studies (Tsai et al., 2011; Wu et al., 2013).

*Gastrodia elata* Blume (GE) is a well-known culinary and medicinal herb belonging to the *Orchidaceae* family and officially listed in the several Chinese ancient books. In the *Compendium of Materia Medica* (本草綱目 běn cǎo gāng mù) written by Li Shizhen in 1596, GE has been used to cure or ameliorate the susceptibility to fright (善驚shàn jīng), an emotion occurs suddenly. *Complete Works of Jingyue* (景岳全書 jǐng yuè quán shū) written by Zhang Jiebin in 1640 quotes that GE possessed effect to stabilize the mind (定志 dìng zhì), which means to cure, recover the mood. The medicinal applications of GE mentioned above are mainly focusing on relieving feelings of anxiety, melancholy and depression. In the modern biochemistry studies, GE has been recognized for its beneficial effects on mental health such as the improvement of learning (Hsieh et al., 2000), as well as its anxiolytic (Jung et al., 2006), and antidepressant-like effects (Chen et al., 2009). Investigations revealed that gastrodin (GAS) is the major active components of GE (Yang et al., 2007), and it has neuroprotective effects (Jiang et al., 2014). Moreover, GAS is able to cross the blood brain barrier (Wang et al., 2007). In our previous study, we showed that the water extract of *Gastrodia elata* Blume (WGE) exerted antidepressant-like effect in the forced swimming test (FST) (Chen et al., 2009); the aim of this study is to investigate the antidepressant-like effects of WGE in UCMS, and to explore its possible molecular mechanisms. We hypothesize that UCMS-induced depressive-like behaviour of animals is caused by the dysregulation of monoamine neurotransmitters and treatment with WGE might ameliorate these behavioural effects. Furthermore, PC12 cells may help to interpret the biochemistry of the antidepressant-like effects of WGE and GAS in rats undergone UCMS.

## 2. Materials and methods

### 2.1. Plant material

WGE was provided by KO DA Pharmaceutical Co., Ltd (Taoyuan, Taiwan). GE was purchased from the Chrysanthemum Village

medicine market, which is a Chinese herb pharmacy in Kunming (Yun-nan, China). The voucher specimen (10411-1307) was authenticated by Dr. Chao-Hsiang Chen and deposited at KO DA Pharmaceutical Company. The specifications and acceptability criterion of GE were based on Taiwan Herbal Pharmacopoeia (Committee on Chinese Medicine and Pharmacy, 2013) using thin layer chromatography (TLC) for analysis with similar retention value (Rf) as compared to reference authentic specimen and high performance liquid chromatography (HPLC) for establishment of fingerprint spectrums. The content of gastrodin was higher than 0.2% (see Supplementary Fig. 1.).

### 2.2. Preparation of the compounds

To obtain WGE, 1 vol. of dried GE rhizome was boiled in 15 vol. of ultrapure water for 1 h. The first extract and residue were obtained after filtering through a 0.1 mm mesh. Then, another 12 vol. of ultrapure water were added to the residue, the mixture was boiled for 1 h, and was further filtered through a 0.1 mm mesh to get the second extract. The two extracts were combined and filtered with 0.05 mm mesh, and the resulting solution was freeze-dried and stored until use. The yield was about 18.9%, and the GAS content in WGE was 3.02% according to the quantification of KO DA (data not shown). GAS (>99.5%) was provided by Wuhan YC Fine Chemical Co. (Wuhan, China). For animal experiments, WGE was dissolved in ultrapure water for oral administration, and for all *in vitro* experiments, compounds were first dissolved in sterilized phosphate-buffered saline (PBS), filtered through a 0.22 mm filter, and then dissolved in the medium to final concentration.

### 2.3. Experimental animals

We used 32 male Sprague-Dawley rats (4-weeks-old, 90–100 g), purchased from BioLASCO Taiwan Co., Ltd. (Ilan, Taiwan). The rats were housed individually in the Animal House Facility, Institute of Food Science and Technology, National Taiwan University at  $22 \pm 2^\circ\text{C}$ , 50–60% humidity, and under a 12 h/12 h light/dark cycle (lights on at 10:00 p.m.). Food (rodent chow diet) and water were provided ad libitum; consumption was recorded every 2 days and the rats were weighed twice a week. All animal protocols were in accordance with the institutional guidelines and were approved by the local ethics committee, the Institutional Animal Care and Use Committee (IACUC) of National Taiwan University, Taiwan (Approval Number: NTU-102-EL-107).

### 2.4. Treatment groups

After 1 week of acclimatization to laboratory conditions, animals were randomly separated into 4 groups (8 rats per group) according to WGE and UCMS treatments: control (CTL), WGE treatment without exposure to UCMS (CW), exposure to UCMS with no drugs administered (UCMS), and treatment with both UCMS and WGE (UW). Ultrapure water (CTL and UCMS groups) or WGE (0.5 g/kg body weight) (CW and UW groups) was administered orally, and the stress paradigm was initiated after another 1 week of acclimatization. Dosage of WGE was established based on our previous studies (Chen et al., 2009; Chen et al., 2016; Lin et al., 2014) and preliminary data for UCMS and we found that WGE (0.5 g/kg body weight) had overall the best antidepressant-like activity.

### 2.5. UCMS procedure

Rats in the CTL and CW group were kept in their home cages undisturbed in a separate animal room during the entire experiment. Rats in both group received only regular cares, food and

water available *ad libitum*, and administrations orally daily. Rats in the UCMS and the UW groups were housed in another animal room, and they were subjected to UCMS for 4 weeks as described previously with some modifications (Dong et al., 2013; Tang et al., 2015). Briefly, the stressors included the following: (1) limited feeding for 1 h; (2) empty drinking bottles for 1 h; (3) deprivation of food and/or water for 24 h; (4) limited space for 1 h; (5) exposure to white noise (80 dB) for 3 h; (6) forced swimming in cold water (8 °C) for 5 min; (7) soiled cage for 18 h; (8) cage tilting at 45° for 8 h; (9) continuous illumination or dark for 24 h; (10) interruption of the light/dark cycle (for 2 h at a time, three times a day). At any time in the day, 2–4 kinds of stressors were introduced randomly and individual stressors were not used more than twice per week, which ensured the unpredictable nature of this model. For all groups, oral administration was performed at 9:00 a.m.

## 2.6. Sucrose preference test

Sucrose preference test was performed to assess anhedonia, one of the core symptoms of depression in humans (Willner, 1997). Before the beginning of UCMS procedure, all rats were given 1% (w/w) sucrose solution for 24 h. Then, both sucrose solution and ultrapure water were provided *ad libitum* for another 24 h. After food and water deprivation for 24 h, the rats were given to both 1% sucrose solution and ultrapure water for 1 h baseline test. Sucrose preference was calculated according to the following formula: sucrose preference (%) = sucrose intake (g)/[sucrose intake (g) + water (g)] × 100%. After the baseline sucrose preference test, the UCMS procedure and oral administration were started. Throughout the whole experiment, sucrose preference test was conducted at 9:00 a.m. every Wednesday after a 24-hour food and water deprivation.

## 2.7. Open field test

Open field test was used to investigate the spontaneous behaviour and locomotor activity of rodents. In the present study, we followed the protocol of Frye and Walf (2002) with slight modifications. The open field apparatus consisted of a 76-cm × 57-cm dark plastic box with 35-cm-high walls, and the box was divided into several rectangular units. Each rat was gently placed in the centre of the box for 30 s for adaptation, and then allowed to freely explore the area for 5 min. After each trial, the apparatus was thoroughly cleaned with 75% ethanol to remove the odour clues left by the previous rat. All behaviours were recorded automatically for 5 min and were analysed blindly by TopScan animal behaviour analytical software (TopScan v2.0 software CleverSys Inc, Reston, VA, USA). The open field test was performed at the end of the experiments.

## 2.8. Tissue sampling

At the end of the experiment, the rats were suffocated by CO<sub>2</sub> and decapitated immediately. The frontal cortex, hippocampus, striatum and amygdala were dissected rapidly and stored at −80 °C until further analysis. The blood was collected during decapitation. The serum was separated using a refrigerated centrifuge at 3000 × g and 4 °C for 20 min, and stored at −80 °C until further analysis.

## 2.9. Measurement of serum corticosterone concentration

As described previously, serum samples were stored at −80 °C until the assay was performed. Corticosterone levels in the serum were measured using a commercial kit (Enzo Life Science Inc.,

Farmingdale, NY, USA) according to the manufacturer's instruction.

## 2.10. Quantification of monoamines

Quantification of monoamines was performed according to the method described in Chen et al. (2009) with some modifications. Briefly, the frontal cortex, hippocampus, striatum, and amygdala were homogenized in 2, 0.8, 1.2, and 0.8 mL of ice-cold homogenizing solution (0.1 N HCl, 0.1 μM ascorbic acid, 1.5 mg/100 mL pargyline, and 50 pg/μL isoproterenol) respectively. Serotonin (5-hydroxytryptophan, 5-HT), 5-hydroxyindoleacetic acid (5-HIAA), dopamine (DA), 3,4-dihydroxyphenylacetic acid (DOPAC), and homovanillic acid (HVA) concentrations in the homogenates were measured with high-performance liquid chromatography (HPLC, L-7100 pump, Hitachi Co. Ltd., Tokyo, Japan) coupled with an electrochemical detector (BAS, West Lafayette, IN, USA) and a column (SPOLAR C18 column, 4.6 × 250 mm, 5 μm, Shiseido Co., Ltd., Tokyo, Japan). The mobile phase contained 20.5 g NaH<sub>2</sub>PO<sub>4</sub>, 185 mg ethylenediaminetetraacetic acid (EDTA), 149.1 mg KCl, 130 mg 1-octanesulfonic acid sodium salt (SOS), and 200 mL methanol in 1 l, adjusted to pH 3.11 using phosphoric acid. The flow rate was 500 μL/min. Conditions of the electrochemical detection: range, 10 nA; Filter, 10 Hz; AppE cell, 0.511 V; 35 °C oven temperature. The chromatographic run time was 30 min, and the retention time of DA, DOPAC, 5-HT, 5-HIAA and HVA were 6.43, 8.69, 12.80, 15.30 and 26.90 min, respectively (see Supplementary Fig. 2). Turnover rates of 5-HT and DA were calculated as 5-HIAA/5-HT and (DOPAC+HVA)/DA, respectively, using the CTL for reference.

## 2.11. Cell culture

PC12 cells were purchased from the Bioresource Collection and Research Centre (BCRC) in Hsinchu, Taiwan. PC12 cells were maintained in Dulbecco's modified Eagle's medium containing 10% horse serum, 5% foetal bovine serum, 100 U/mL penicillin, and 100 μg/mL streptomycin, and the cells were cultured in an atmosphere of 5% CO<sub>2</sub> at 37 °C.

## 2.12. MAO-A inhibitory assay

The MAO-A inhibitory assay was performed using a commercial kit (Promega Co., Madison, WI, USA) according to the manufacturer's instruction with slight modifications (Matsuda et al., 2012; Wells et al., 2010). The bioassay herein comprised a direct inhibitory and an indirect inhibitory assay, and the MAO-A proteins used in both assays were isolated from PC12 cells. In the direct inhibitory assay, PC12 cells were seeded at a density of 1 × 10<sup>6</sup> cells/mL in a 10-cm dish and were incubated for 24 h. Then the medium was removed, the cells were washed twice with PBS, 400 μL RIPA buffer was added, and the cells were kept on ice for 5 min to lyse the cells. The extract was centrifuged for 10 min at 14,000 × g in a cold microfuge to obtain the lysates, which contained MAO-A. WGE (0.5 mg/mL) and gastrodin (15, 30, and 60 μg/mL) were added to the lysates to assess their inhibitory effects on MAO activity, and clorgyline (10 μM) was used as a reference inhibitor of MAO-A. In the indirect inhibitory assay, PC12 cells were seeded at a density of 3 × 10<sup>5</sup> cells/well in 6-well plates and were incubated for 24 h. After that, all compounds were co-incubated with the cells for 48 h, and then, lysates for analysis of MAO-A activity were obtained according to a protocol described previously. Activity of MAO-A is shown in relative luminescence units (RLU) per mg protein using Bradford assay (Bradford, 1976) for normalization and the CTL for reference.

### 2.13. Western blot analysis

PC12 cells were seeded at a density of  $1 \times 10^5$  cells/mL in a 10-cm dish and were incubated for 24 h, and then co-incubation with WGE (0.5 mg/mL), GAS (15, 30 and 60  $\mu$ g/mL) and clorgyline (10  $\mu$ M, for MAO-A only) for another 48 h to examine the expression of proteins regulating the level of monoamines. The methods of SDS polyacrylamide gel electrophoresis (SDS-PAGE) and western blot were according to our previous study (Lin et al., 2014). The following primary antibodies were used: anti-MAO-A (Abcam Inc., Cambridge, MA, USA), anti-TH (Merck Millipore, Darmstadt, Germany) and anti-glyceraldehyde 3-phosphate dehydrogenase (GAPDH, Genetex, Irvine, CA, USA).

### 2.14. Statistical analyses

Data are presented as mean  $\pm$  standard deviation (SD). Statistical analysis was performed using the unpaired Student's *t*-test for the CTL and CW groups *in vivo*. *P* values less than 0.05 indicated significantly differences from the CTL group (# denotes  $p < 0.05$ , ## denotes  $p < 0.01$ , and ### denotes  $p < 0.001$ ). As for the CTL, UCMS and UW groups *in vivo*; and all groups *in vitro*, we used the one-way analysis of variance test (ANOVA) with Newman-Keuls post-hoc test. *P* values less than 0.05 indicated significantly differences from the UCMS group *in vivo* and CTL *in vitro* (\* denotes  $p < 0.05$ , \*\* denotes  $p < 0.01$ , and \*\*\* denotes  $p < 0.001$ ).

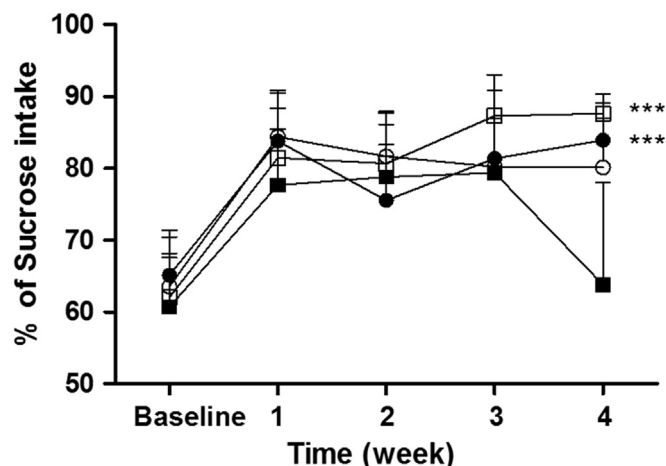
## 3. Results

### 3.1. Effects of WGE on body weight and daily intakes

Body weight at baseline was measured before the UCMS paradigm was initiated and weight was measured weekly until the end of the experiment. As shown in Fig. 1(A), there were no significant differences in body weights between the groups at baseline, but the body weight of rats exposed to stress (UCMS) was significantly lower than that in the CTL group from week 1 to week 4. The body weight of rats in the UW group was higher than that in the UCMS group at week 4, and the body weight of rats in the CW group was higher than that in the CTL group at week 3 and 4. Food and water intakes were measured every 2 days, and the averages of food and water consumption are shown in Fig. 1(B) and (C), respectively. In the UCMS group, both food and water intakes decreased at week 4 compared with the CTL group, but the UW treatment prevented the decreases induced by the UCMS treatment. CW treatment caused a slight increase in the food intake compared with CTL treatment at week 3 (Fig. 1(B)).

### 3.2. Effects of WGE on sucrose preference test

As shown in Fig. 2, sucrose preference did not differ between the groups at baseline. After 4 weeks of UCMS exposure, the sucrose preference in the UCMS



**Fig. 2.** Effects of the water extract of *Gastrodia elata* Blume (WGE) on sucrose preference during the experiment. CTL (●), control; CW (○), control plus WGE treatment (0.5 g/kg body weight); UCMS (■), unpredictable chronic mild stress; UW (□), UCMS plus WGE treatment (0.5 g/kg body weight). Data are shown as mean  $\pm$  standard deviation (SD,  $n=8$ ). Statistical analyses were performed using the unpaired Student's *t*-test for the CTL and CW group, and the one-way ANOVA with Newman-Keuls post-hoc test (\*\*\* $p < 0.001$ : significantly different from the UCMS group).

group dramatically decreased compared to that in the CTL group, while the sucrose preference remained unaffected in the UW group.

### 3.3. Effects of WGE in the open field test

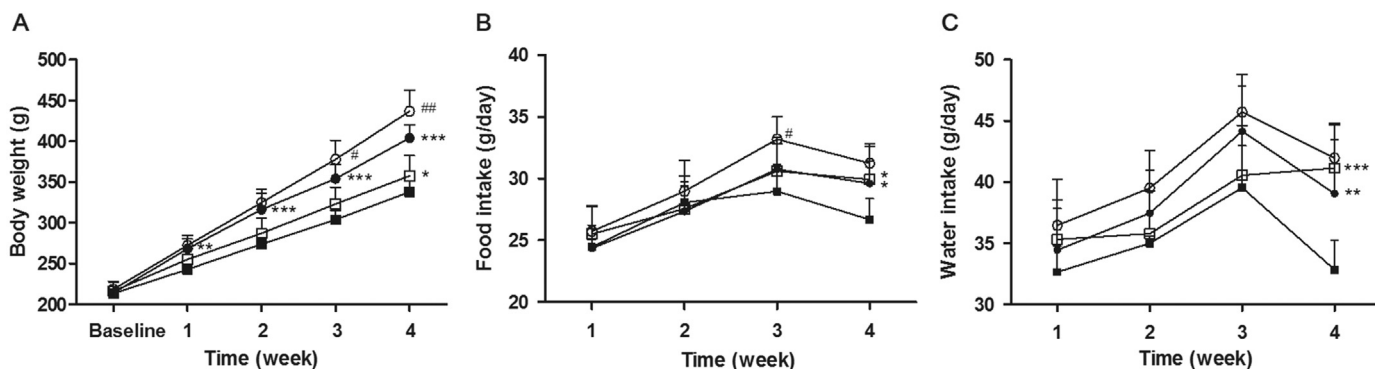
In order to confirm the onset of depression-like behaviour at week 4, open field test was performed. Several indicators of depression-like behaviour (total travelled distance; number of crossing, rearing and grooming behaviours) are shown in Fig. 3. All behavioural indicators decreased in the UCMS group compared to that in the CTL group, but treatment with WGE (UW) reversed these abnormal behaviours.

### 3.4. Effects of WGE on serum corticosterone levels

As shown in Fig. 4, serum corticosterone levels in the UCMS group were significantly elevated compared to that in the CTL group, while UW treatment prevented this UCMS-induced elevation. There was no significant difference in serum corticosterone levels between the CW and CTL groups.

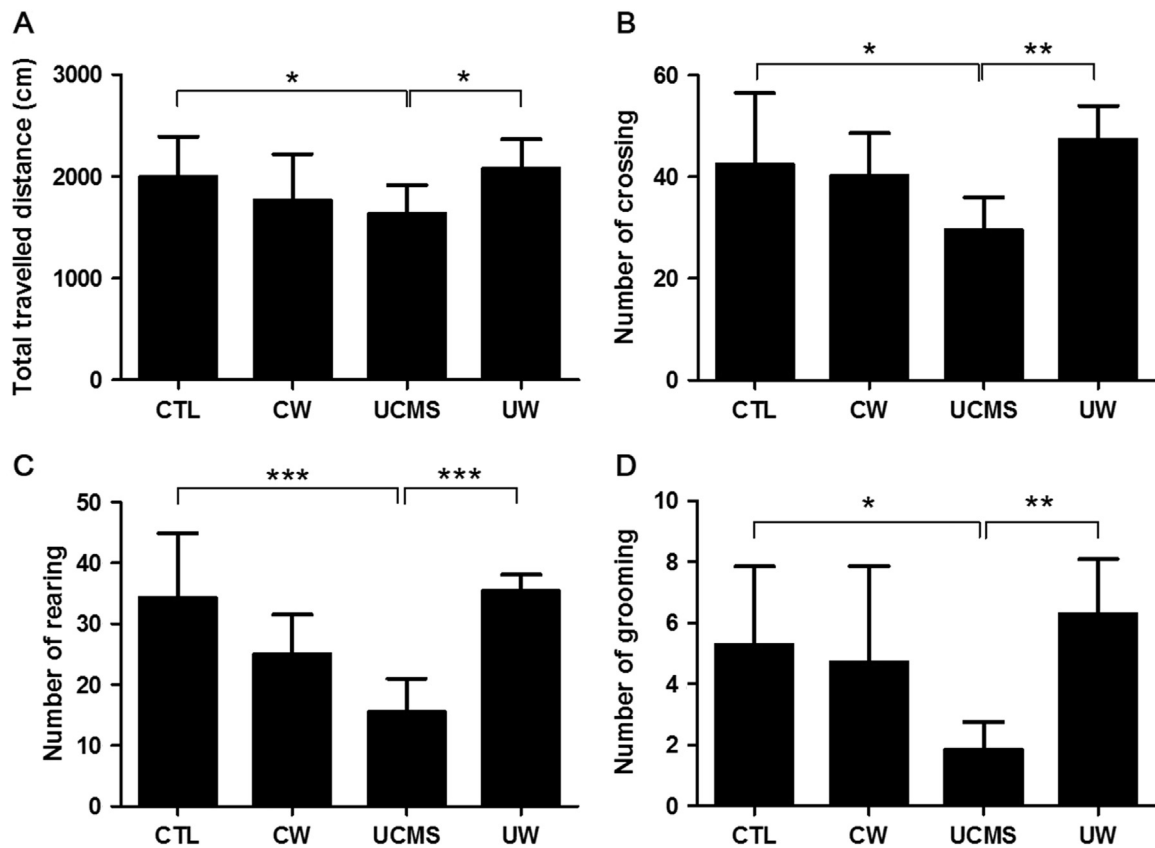
### 3.5. Effects of WGE on turnover rates of monoamines in the selected brain areas

Fig. 5 shows the turnover rates of 5-HT and DA in the frontal cortex, hippocampus, striatum, and amygdala. Turnover rates of 5-HT and DA increased in the UCMS group compared with that in the CTL group in all the investigated brain areas, while the UW treatment ameliorated these elevated turnover rates of both 5-HT and DA induced by UCMS. In addition, the turnover rate of 5-HT in the frontal

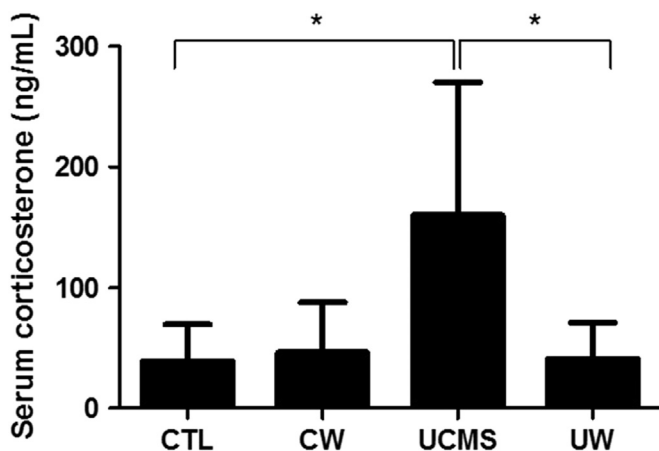


**Fig. 1.** Effects of the water extract of *Gastrodia elata* Blume (WGE) on body weight (A), food intake (B) and water intake (C) during the experiment. CTL (●), control; CW (○), control plus WGE treatment (0.5 g/kg body weight); UCMS (■), unpredictable chronic mild stress; UW (□), UCMS plus WGE treatment (0.5 g/kg body weight). Data are shown as mean  $\pm$  standard deviation (SD,  $n=8$ ). Statistical analyses were performed using the unpaired Student's *t*-test for the CTL and CW group (#  $p < 0.05$  and ##  $p < 0.01$ : significantly different from the CTL group), and the one-way ANOVA with Newman-Keuls post-hoc test (\* $p < 0.05$ ; \*\* $p < 0.01$  and \*\*\* $p < 0.001$ : significantly different from the UCMS group).





**Fig. 3.** Effects of the water extract of *Gastrodia elata* Blume (WGE) on total travelled distance (A), the number of crossings (B), the number of rearing behaviours (C), and the number of grooming behaviours (D) in the open field test. CTL, control; CW, control plus WGE treatment (0.5 g/kg body weight); UCMS, unpredictable chronic mild stress; UW, UCMS plus WGE treatment (0.5 g/kg body weight). Data are shown as mean  $\pm$  standard deviation (SD,  $n=8$ ). Statistical analyses were performed using the unpaired Student's *t*-test for the CTL and CW group, and the one-way ANOVA with Newman-Keuls post-hoc test (\* $p < 0.05$ ; \*\* $p < 0.01$  and \*\*\* $p < 0.001$ : significantly different from the UCMS group).



**Fig. 4.** Effects of the water extract of *Gastrodia elata* Blume (WGE) on serum corticosterone levels. CTL, control; CW, control plus WGE treatment (0.5 g/kg body weight); UCMS, unpredictable chronic mild stress; UW, UCMS plus WGE treatment (0.5 g/kg body weight). Data are shown as mean  $\pm$  standard deviation (SD,  $n=8$ ). Statistical analyses were performed using the unpaired Student's *t*-test for the CTL and CW group, and the one-way ANOVA with Newman-Keuls post-hoc test (\* $p < 0.05$ : significantly different from the UCMS group).

cortex of rats in the CW group decreased compared to that in the CTL group (Fig. 5A). However, the turnover rate of DA slightly increased in the hippocampus (Fig. 5F) and striatum (Fig. 5G) of the CW group compared to that in the CTL group.

### 3.6. Effects of WGE and GAS on MAO-A activity

To investigate whether WGE alleviated the high turnover rates of monoamines

by inhibiting MAO-A, a MAO-A inhibitory assay was performed. WGE (0.5 mg/mL) and its major compound, GAS (15, 30, and 60  $\mu$ g/mL), did not affect cell viability 48 h after co-incubation (see Supplementary Fig. 3). As shown in Fig. 6(A), WGE and a low dose of GAS had no inhibitory effect on MAO-A activity compared with the CTL treatment, while a medium and high dose of GAS markedly inhibited MAO-A. Clorgyline was used as a positive control. Fig. 6(B) shows that all 48-h treatments had indirect inhibitory effects on MAO-A, and a trend of dose-dependent effect of MAO-A inhibition of GAS was observed.

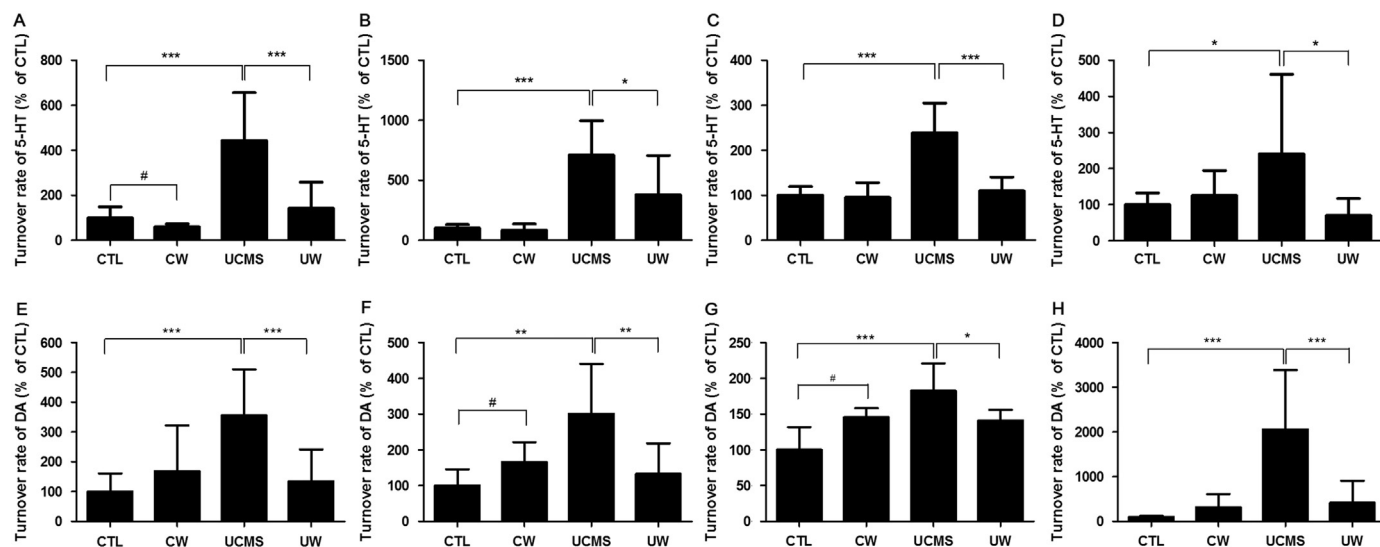
### 3.7. Effects of WGE and GAS on the expressions of MAO-A and TH

Fig. 7(A) and (B) show the protein expression of MAO-A and TH, respectively. WGE and all concentrations of GAS repressed the expression of MAO-A and increased the expression of TH; the trends of dose-dependent effects of GAS on both experiments were observed.

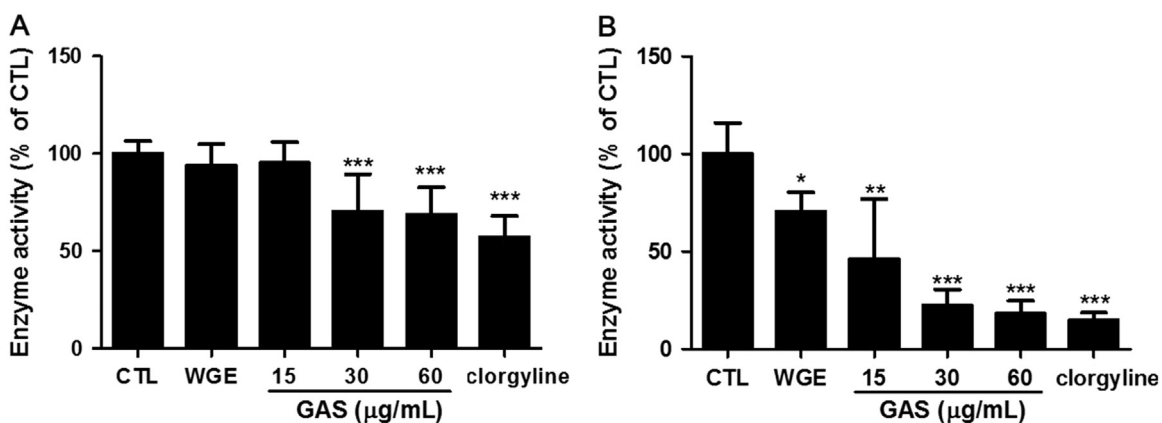
## 4. Discussion

In the present study, we studied the effects of WGE in the animal model of UCMS-induced depression. The results indicated that 4-week-long UCMS induced significant depression-like behaviours, which was characterized by growth retardation, decreased food and water intakes, anhedonia, and decreased exploring activities and spontaneous behaviours. Furthermore, UCMS elevated corticosterone levels and increased the turnover rates of 5-HT and DA. Administration of WGE significantly ameliorated the abnormal behaviours and biochemical changes. Our significant finding was that similar to WGE, GAS directly inhibited both MAO-A activity and protein expression. In addition, WGE and GAS increased the protein expression of TH.

There are several significances between FST and UCMS models.



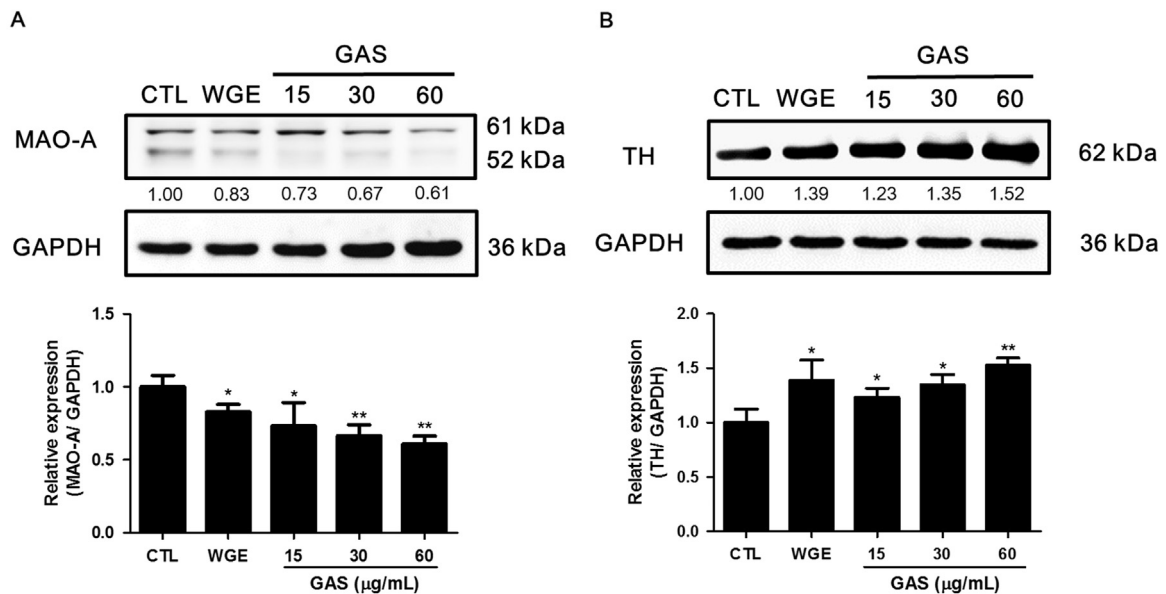
**Fig. 5.** Effects of the water extract of *Gastrodia elata* Blume (WGE) on the turnover rate of serotonin (5-HT) in the frontal cortex (A), hippocampus (B), striatum (C), and amygdala (D) and on the turnover rate of dopamine (DA) in the frontal cortex (E), hippocampus (F), striatum (G) and amygdala (H). CTL, control; CW, control plus WGE treatment (0.5 g/kg body weight); UCMS, unpredictable chronic mild stress; UW, UCMS plus WGE treatment (0.5 g/kg body weight). Data are shown as mean  $\pm$  standard deviation (SD,  $n=8$ ). Statistical analyses were performed using the unpaired Student's *t*-test for the CTL and CW group (#  $p < 0.05$ : significantly different from the CTL group), and the one-way ANOVA with Newman-Keuls post-hoc test (\* $p < 0.05$ ; \*\* $p < 0.01$  and \*\*\* $p < 0.001$ : significantly different from the UCMS group).



**Fig. 6.** Direct (A) and indirect inhibitory effects (B) of the water extract of *Gastrodia elata* Blume (WGE) and gastrodin on monoamine oxidase A (MAO-A) activity. CTL, control; WGE, 0.5 mg/mL; GAS, 15, 30 and 60  $\mu\text{g/mL}$  gastrodin; cloglyline, 10  $\mu\text{M}$ . Data are shown as mean  $\pm$  standard deviation (SD,  $n=3$ ). Statistical analysis was performed using the one-way ANOVA with Newman-Keuls post-hoc test (\* $p < 0.05$ , \*\* $p < 0.01$ , and \*\*\* $p < 0.001$ : significantly different from the CTL group).

The assessment of depressive-like behaviours in FST is based on the observation of the immobility time when rodents are placed in an inescapable, water-containing cylinder. The immobile posture is therefore interpreted as behavioural despair and can be altered by treatment with clinical antidepressants (Cryan and Slattery, 2007). FST is a high throughput, ease to use and common test to evaluate the efficacy of antidepressants. However, a major criticism is that it does not reflect the situation of human depression since the stress used in FST is an acute and tremendous intervention (Slattery and Cryan, 2014). In comparison with FST, the model UCMS introduces several stressors, reduces their severity to a moderate level, extends the duration of the stress paradigm and makes them unpredictable. These result in many behavioural abnormalities such as decreases in daily intakes, locomotor activity and the onset of the key character for depressive-like phenomenon, anhedonia. Most of these behavioural defects combining with endocrinal and neuronal changes are found similar in depression patients and able to be improved by treatment with chronic antidepressants (Willner, 1997). Due to its good ecological validity, it is regarded as one of the most useful animal model to study the outcomes of antidepressants.

Consistent with the results of previous study, we observed a decrease in body weight and food intakes after UCMS (Su et al., 2014), while WGE ameliorated these changes. These benefits may be the results of an increased or preserved appetite based on our results. Decrease of sucrose preference, a form of anhedonia, is considered as an indicator for depressive-like behaviour. Anhedonia is defined as the decrease in responsiveness to natural reward by reducing the intake of palatable food and sweet drinking fluids (Willner, 1997). Our result demonstrated that WGE prevented the decrease of sucrose preference induced by UCMS protocol, suggesting the antidepressant-like effect of WGE, which is in accordance with our previous findings using the FST model (Chen et al., 2009). The open field test investigates the behaviour of murine in a novel environments (Hall, 1934). In an emotionally affected animal, locomotor activity and the number of rearing, grooming and other spontaneous behaviours are decreased (Katz, 1981). After a 4-week-long UCMS treatment, the rats became hypoactive: the total travelled distance decreased, novelty seeking behaviour (crossing and rearing) weakened, and the number of grooming actions reduced, suggesting a depression-like phenotype in line with the results of a previous study (Cai et al., 2015).



**Fig. 7.** Effects of the water extract of *Gastrodia elata* Blume (WGE) and gastrodin on the expression of monoamine oxidase A (MAO-A, panel A) and tyrosine hydroxylase (TH, panel B). CTL, control; WGE, 0.5 mg/mL; GAS, 15, 30 and 60 µg/mL gastrodin. Data are shown as mean  $\pm$  standard deviation (SD,  $n=3$ ). Statistical analysis was performed using the one-way ANOVA with Newman-Keuls post-hoc test (\* $p < 0.05$  and \*\* $p < 0.01$ : significantly different from the CTL group).

However, these abnormal behaviours could be reversed by a treatment with WGE. Additionally, WGE treatment alone (CW group) had no effects on any of the behavioural indicators, demonstrating that the improvements in the open field test were not due to the effects of WGE on locomotor activity.

There is a strong evidence that links stress, the hypothalamic-pituitary-adrenal (HPA) axis, and depression (Dinan, 1994). The HPA axis is an endocrine system regulating the response to stress. Upon exposure to a stressful condition, corticotropin-releasing factor (CRF) is released from hypothalamus, stimulating the pituitary gland to secrete adrenocorticotrophic hormone (ACTH), and thus, glucocorticoid molecules (cortisol in humans and corticosterone in murine) are released by the adrenal cortex in respond to the stress encountered. In general, the HPA axis enables a negative feedback mechanism so that glucocorticoid levels in the circulatory system remain within a normal range. However, long-term stress exposure disturbs this feedback, leading to neuronal injury or death (Sapolsky, 1992). Clinical reports indicate that depression is accompanied by the atrophy of hippocampus in patients who had high blood cortisol levels (Nestler et al., 2002), suggesting that the abnormal hyperactivity of the HPA axis might induce depression. Accordingly, our result indicated that a 4-weeks-long UCMS protocol significantly elevated of serum corticosterone levels, while a long-term treatment with WGE ameliorated this increase and exhibited antidepressant-like effects similar to certain antidepressants (Mason and Pariente, 2006).

The monoamine deficiency theory was proposed in 1965 (Gestblom et al., 2008), and a study has indicated that chronic stress exposure reduces the amounts of monoamine neurotransmitters 5-HT and DA in the brains of animals exposed to UCMS (Bhutani et al., 2009). Moreover, it has been reported that the decrease of 5-HT and DA levels is a key pathological sign of depression onset (Holmes et al., 2003). Animal studies even show decreases of cerebral 5-HT and DA in the rats exposed to the FST (Chen et al., 2009) and the tail suspension test (TST) (Hao et al., 2013). In the present study, the turnover rates of 5-HT and DA were significantly increased in all brain areas investigated after the UCMS procedure, suggesting the impaired utilization of monoamines. These results might be partially explained with the decreased levels of both monoamines in certain brain areas (see Supplementary Table 1 and 2), also shown

by a previous study (Su et al., 2014). Our data demonstrated that the oral administration of WGE reversed the higher turnover rates induced by UCMS. According to our findings, we imply that the inhibition of MAO-A activity may ameliorate the higher turnover rates. Therefore we conducted an experiment using PC12 cells as an *in vitro* model to investigate whether WGE and GAS inhibited MAO-A. As described previously, GAS content in WGE was 3.02%, and therefore, 0.5 mg/mL WGE contains 15 µg/mL GAS. The direct inhibitory assay revealed that 30 and 60 µg/mL GAS inhibited the activity of MAO-A, and WGE and all doses of GAS used in this experiment had an indirect inhibitory effect and decreased the protein expression after 48 h of co-incubation. These results support that WGE inhibits MAO-A in our neurotransmitters related experiment. Another study showed that GE inhibits MAO activity and has beneficial effects on learning and memory (Niu et al., 2004).

Recent studies indicated that chronic stress increases both MAO-A activity (Lin et al., 2005) and protein expression (Duncan et al., 2012). Prolonged elevation of glucocorticoids upregulates the expression of Krüppel-like factor 11, a positive transcription factor of MAO-A, resulting in high translation rate of MAO-A. Additionally, another *in vitro* study supports the upstream and downstream correlation between glucocorticoid levels and MAO-A protein expression (Grunewald et al., 2012). These findings suggesting that in addition to inhibiting MAO-A activity, WGE normalizes the HPA axis and downregulates the translation of MAO-A.

Based on the inhibitory effect of WGE on MAO-A activity, the 5-HT turnover rate found in the frontal cortex of rats in the CW group is a reasonable result. However, the relatively high turnover rate of DA in the hippocampus and striatum of rats in the CW group compared with those in the CTL group are more interesting. Amounts of monoamines and their metabolic products can alter the turnover rate, namely the ratio of synthesis and degradation. We found an increase on TH, a key enzyme responsible for DA synthesis, after WGE and GAS treatment for 48 h. These results, the upregulation of TH upon WGE and GAS treatment, were similar to the results of other studies (Kumar et al., 2013; Shin et al., 2011). One possible explanation for higher turnover rate in the CW group might be the compensatory effect of monoamine homeostasis in the brain after the MAO-A inhibition and upregulation of

TH levels of WGE, both could increase the amounts of DA in the cerebrum. The compensatory effect is commonly found in patients with depression who take SSRI antidepressants (Meyer et al., 2009). Another possible reason might be that the elevated activity of catechol-O-methyl transferase (Bhutani et al., 2009; Maura et al., 1978) increases DA metabolism after the inhibition of MAO-A. However, all behavioural data of the CW group remained normal compared with the CTL group, and no other side effects were observed during the experiments. Our study suggests that using traditional herbal medicine, WGE might be a safe alternative to complementarily treat depression.

## 5. Conclusions

In summary, the present study indicates that WGE has antidepressant-like effects, which might be mediated, at least in part, by the regulation of monoamines, more precisely by the down-regulation of MAO-A and the upregulation of TH to increase the half-life and thus the amount of monoamines in the brains. These findings revealed that both WGE and GAS, its major compound, have potential for depression treatment as complementary antidepressants or adjuvants.

## Conflict of interest

None.

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## Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.jep.2016.04.032>.

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